

Geospatial Simulation of Fire Ecology with DeepEarth 🔥

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- Fuel moisture content, the mass of water contained within vegetation in relation to the dry mass.
- FMC is usually separated into dead (DFMC) and live (LFMC) components.
- DFMC is empirically determined from different variables.
- LFMC is difficult to estimate because living plants have a variety of drought adaptation strategies.



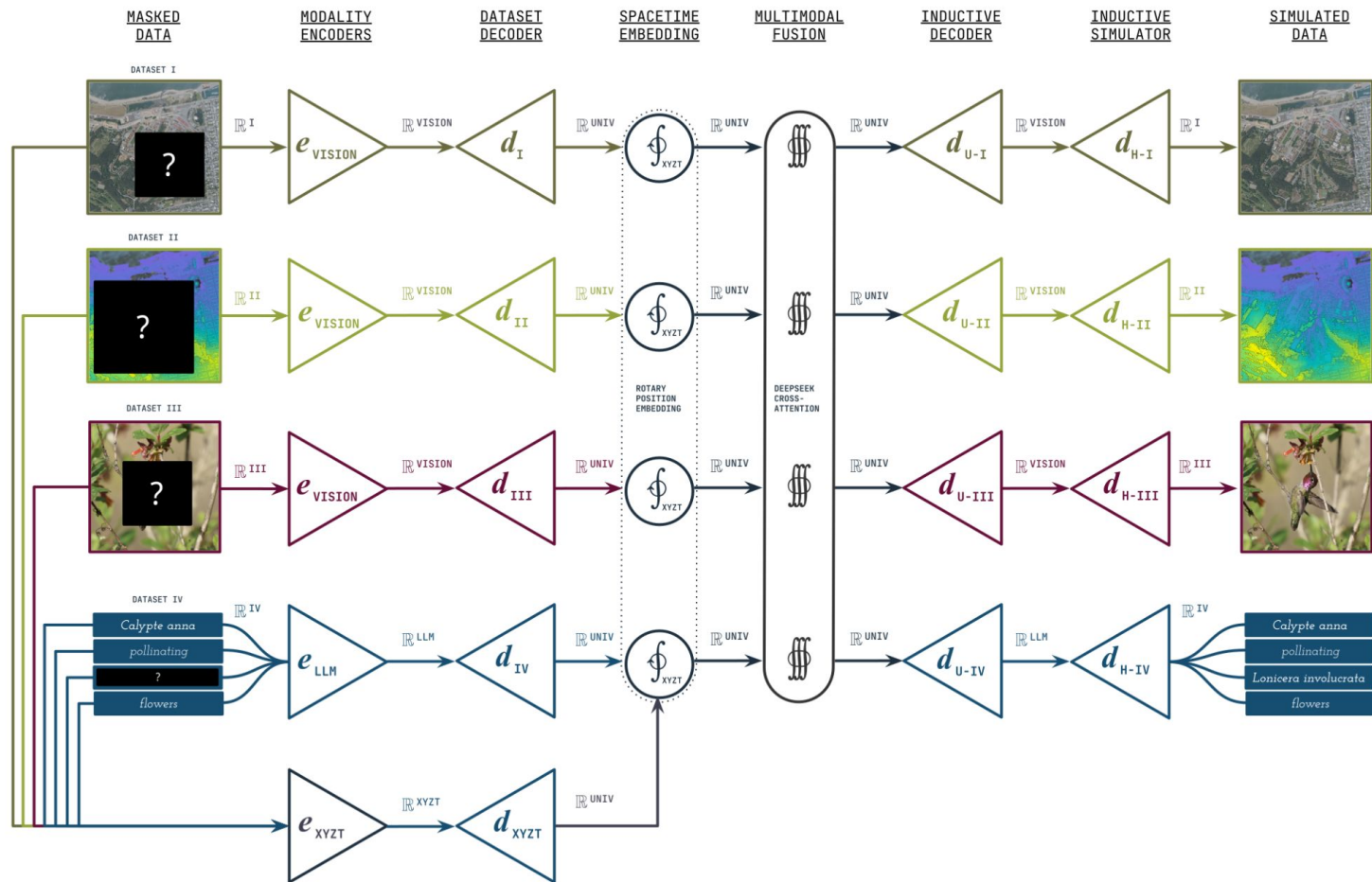
Wildfire burning in the Kaibab National Forest, Arizona, United States, in 2020 (Wikipedia)

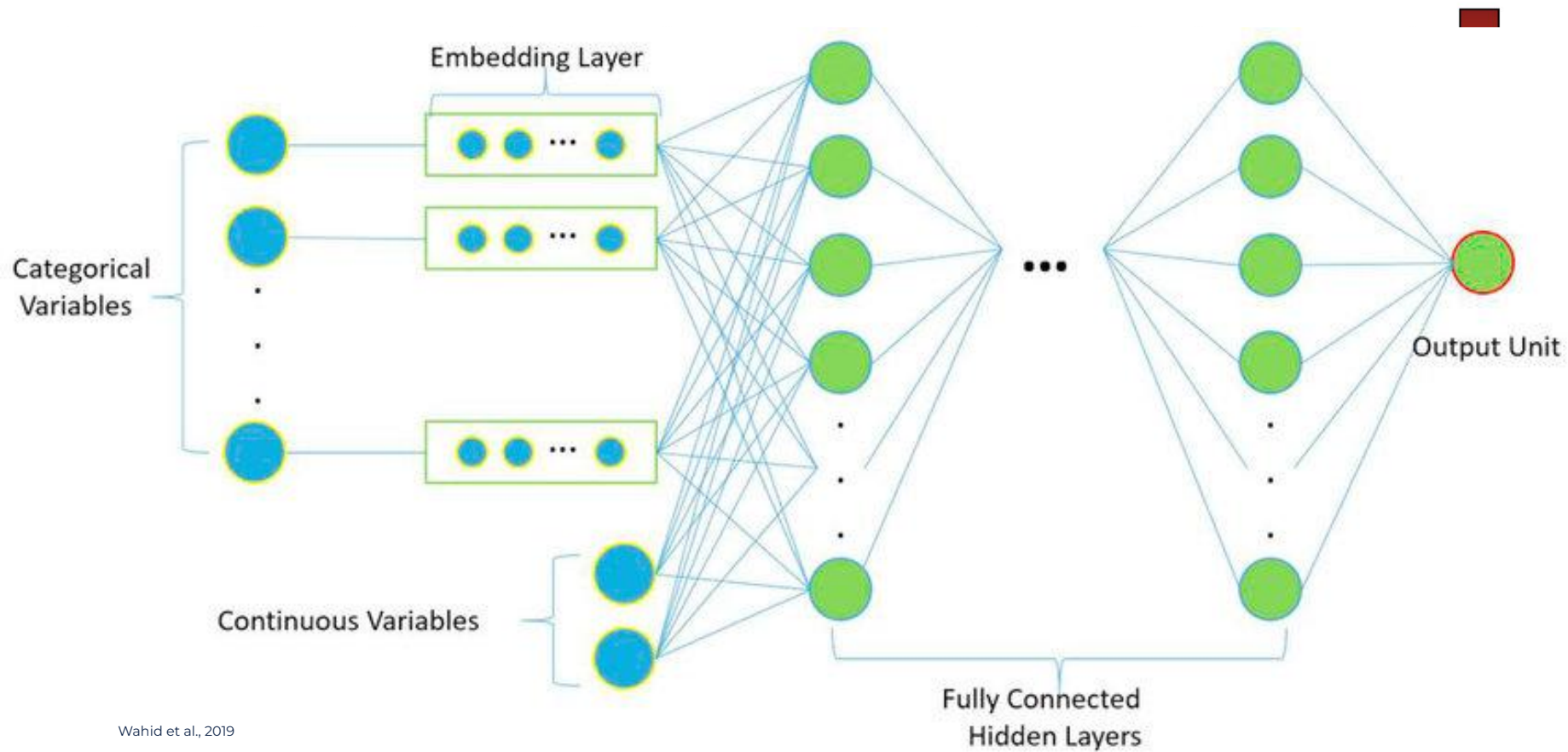
- **What We Did**

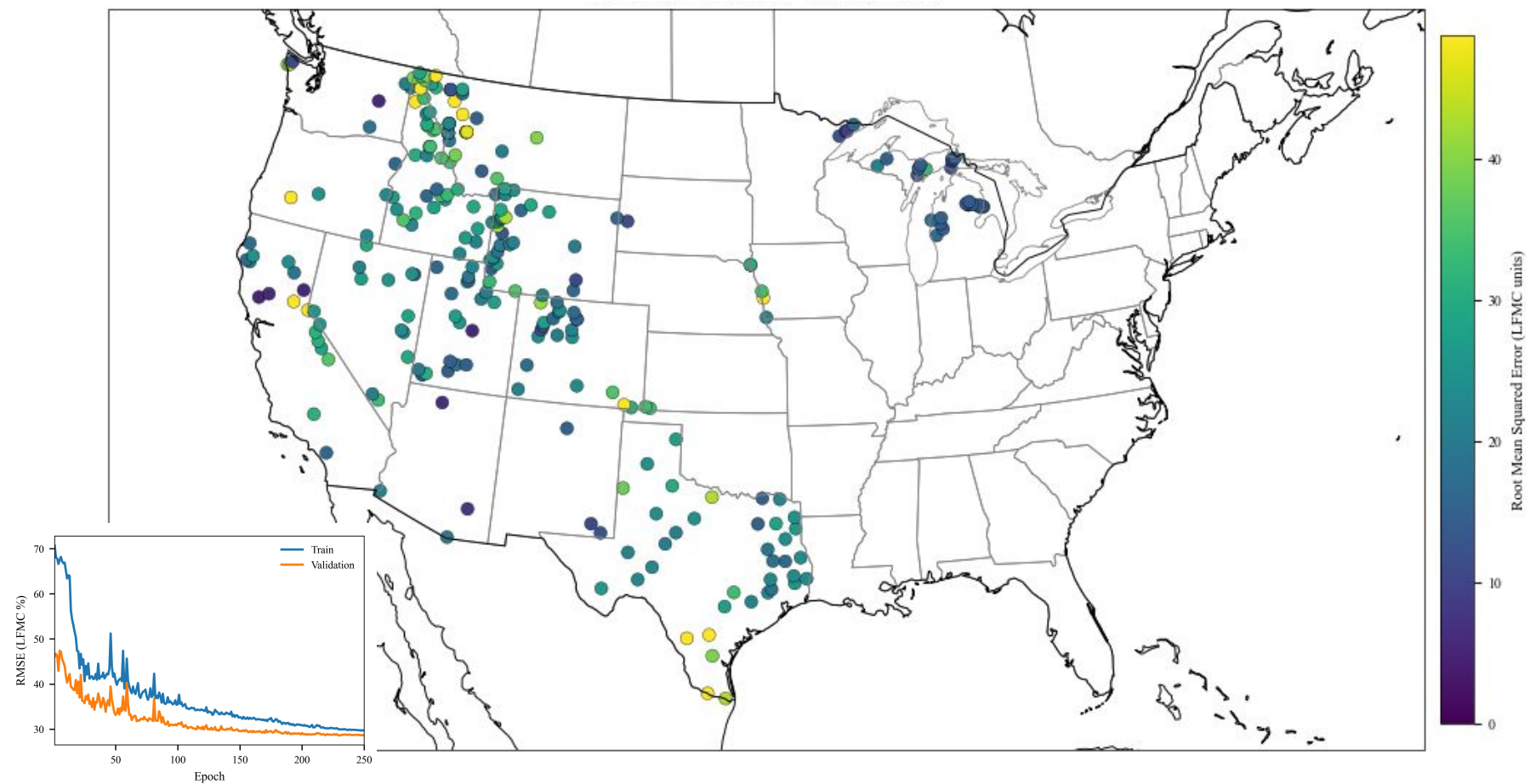
We aim to develop an AI-based geospatial modeling framework to predict LFMC across diverse plant species under complex geospatial, meteorological, and ecological variables.

Each of us focused on a specific section of the project, contributing to the overall system design and research direction.









Generative Resilience

Fire Risk & Fire Resilience

Fire risk – Species' contribution to ignition, spread, or intensity of fire through traits (e.g., flammability, volatile compounds, crown structure, litter type) or indirect effects (e.g., ember transport).

Fire resilience – Individual's ability to survive and recover after fire via traits such as thick bark, protected buds, root crown regeneration, epicormic sprouting.



Pipeline Diagram

Geospatial Sampling —
Coordinates from target areas

Biodiversity Data — GBIF
occurrence records per location

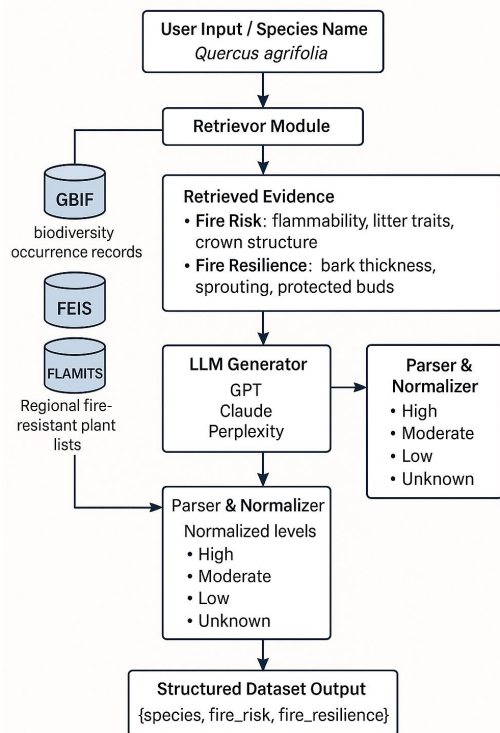
LLM Evaluation — GPT,
Perplexity, Claude prompt with
risk/resilience definitions

Parsing & Normalization —
Levels: High, Moderate, Low,
Unknown

Dataset — Structured {species,
risk, resilience} table

Generative Resilience

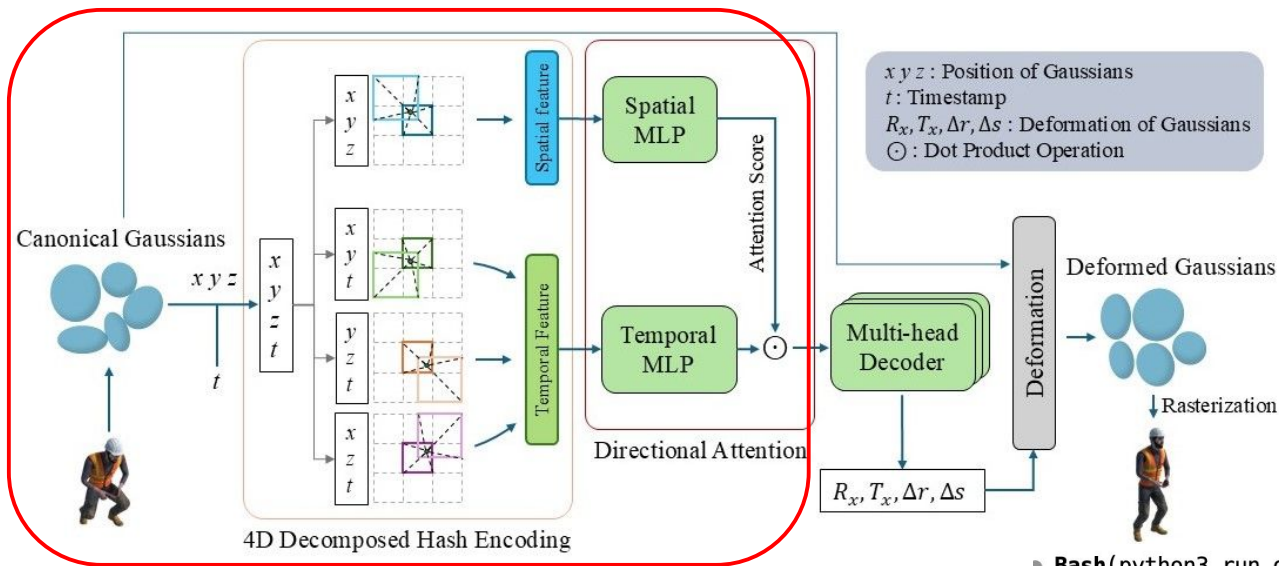
Fire Risk & Fire Resilience



Findings Summary (with small bar chart placeholder)

- **No Data:** Many species have no fire-related info in LLMs or references.
- **Conflicting Data:** LLMs disagree, or LLM vs. literature mismatch.
- **Mismatch Cases:** Even consistent literature can be contradicted by LLM outputs.
- **Partial Matches:** Some species align across all sources.

Grid4D - Predicting LFMC% via decomposing 4D (x,y,z,t) inputs



Globe-LFMC-2.0 data

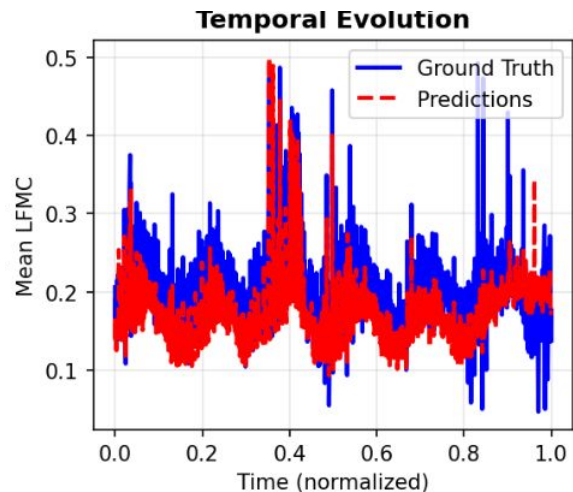
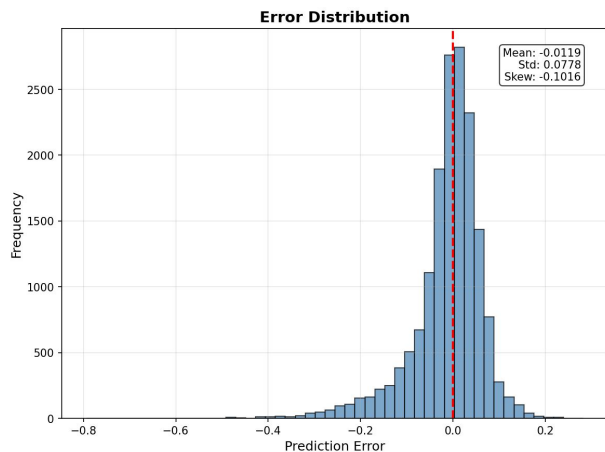
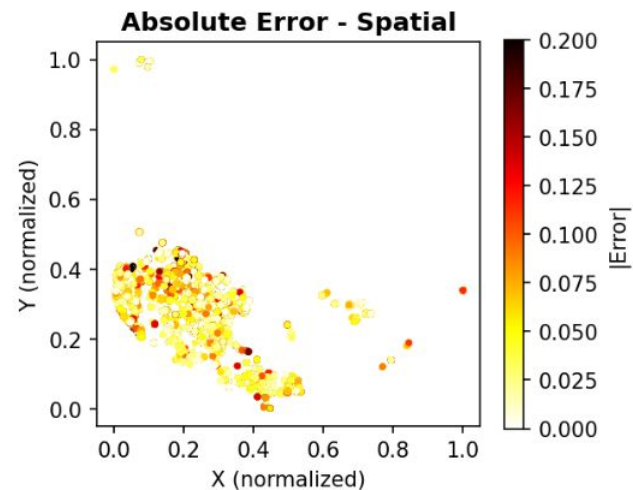
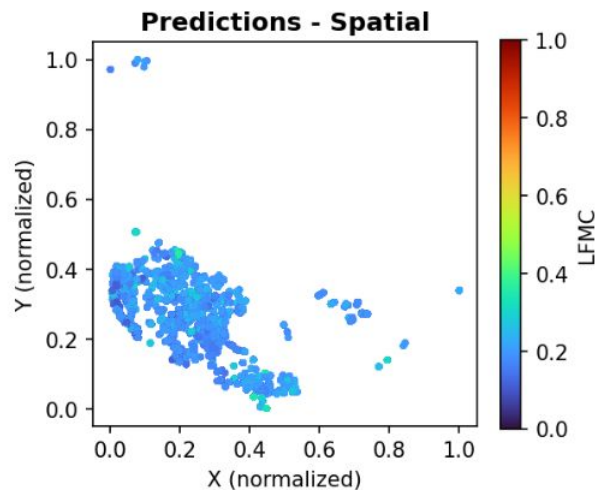
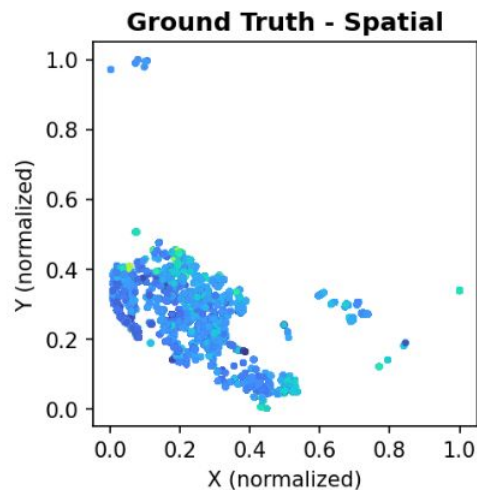
- 2017-2023, CONUS
- Total of 89,702 samples
- (lat, lon, elevation, time stamps)

Grid4D

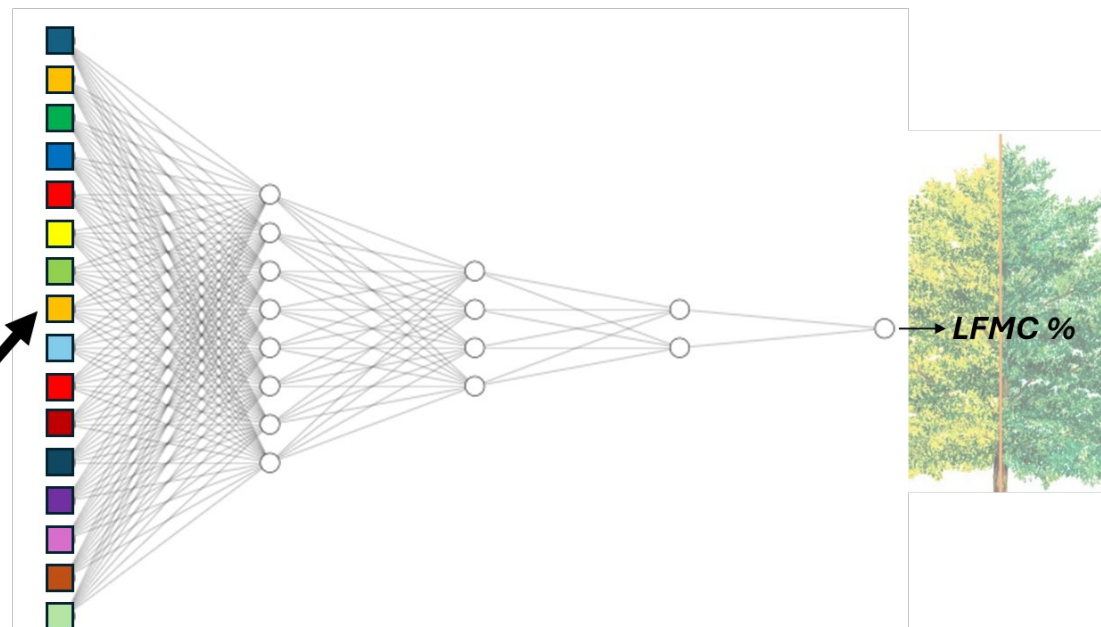
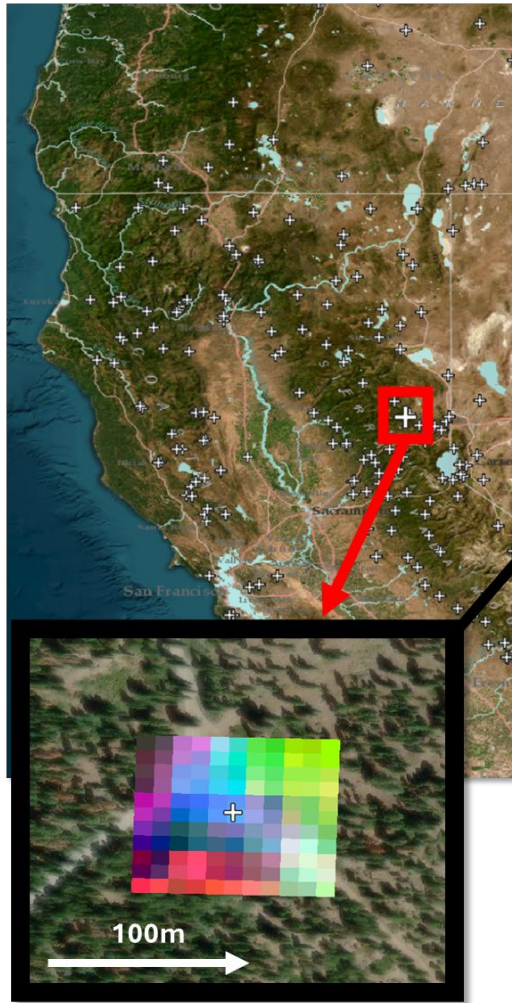
- Decomposed hash encodings over (x,y,z), (x,y,t), (x,z,t), (y,z,t)
- 128D vector -> MLP -> LFMC

```

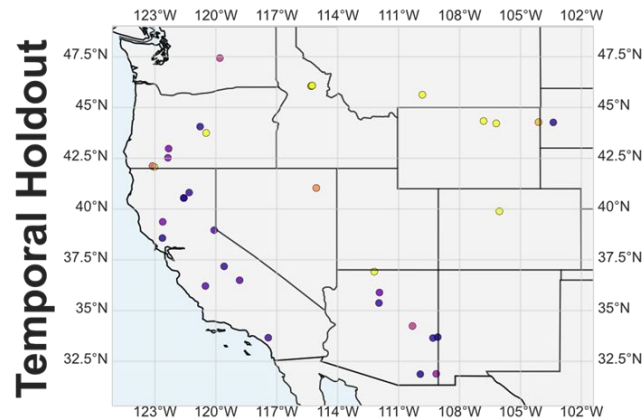
bash$ python3 run_grid4d_training.py
└─ Trainable Parameters: 67,315,841
   └─ Encoder: 67,108,864
      └─ Attention: 99,072
         └─ MLP: 107,905
    
```



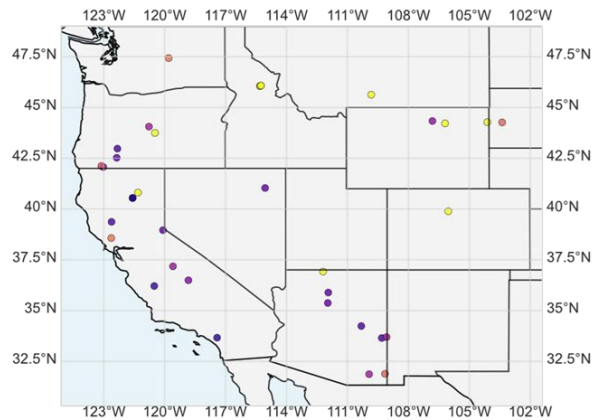
lfmc_normalized	predictions	error
0.218436873747495	0.23639151	0.01795464079222364
0.1743486973947896	0.1847515	0.010402798324166407
0.1683366733466934	0.1603983	-0.007938368884260527
0.1743486973947896	0.14664014	-0.027708560556830297
0.3386773547094188	0.35864738	0.01997002158948563
0.2745490981963928	0.26700082	-0.007548273983364828
0.2605210420841683	0.216583	-0.043938043450783615
0.2224448897795591	0.17144196	-0.0510029324250135
0.1182364729458917	0.15426373	0.03602726187161319
0.1703406813627254	0.1397073	-0.0306333842760097
0.2144288577154308	0.13175818	-0.0826706744745403
0.1282565130260521	0.12654011	-0.001716403510861969
0.2124248496993988	0.2881433	0.07571845727119753



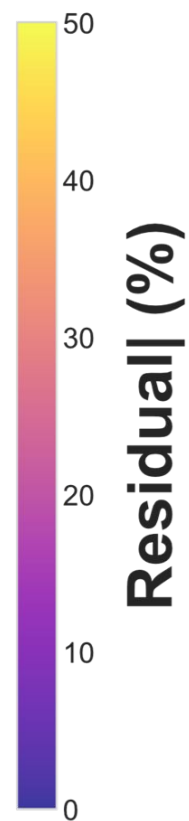
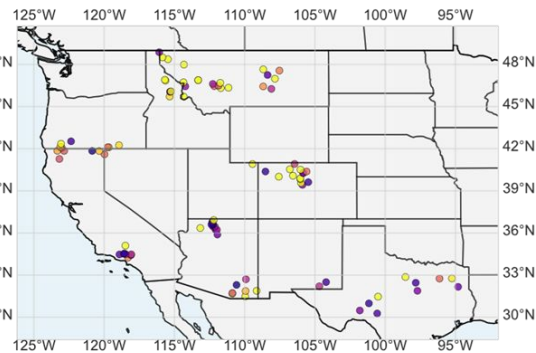
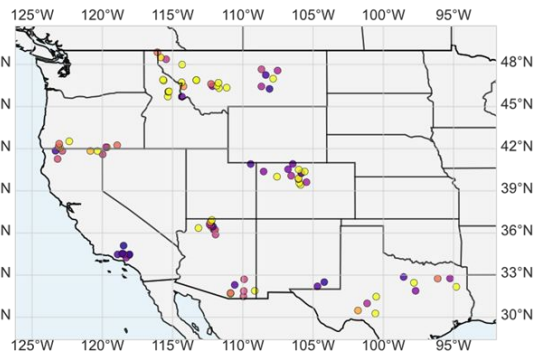
CNN



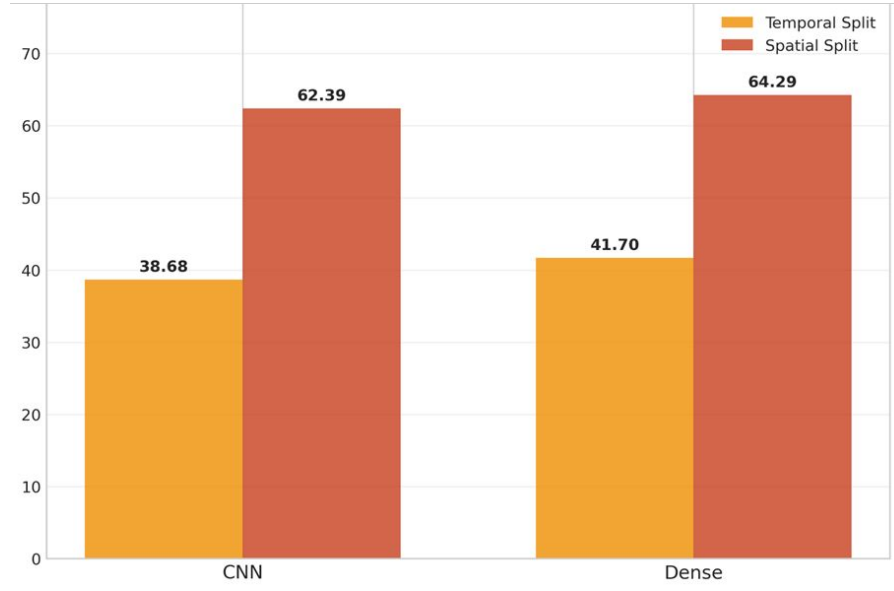
MLP



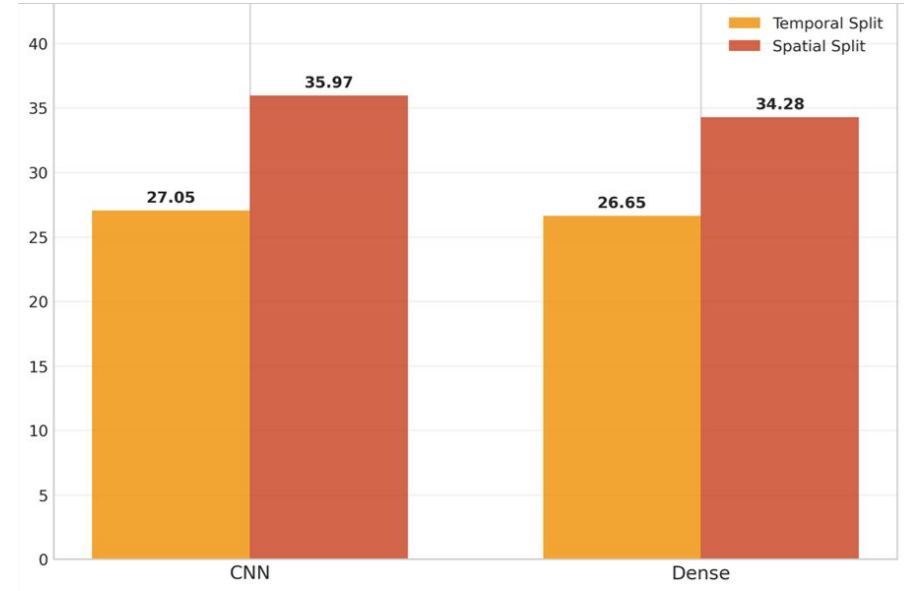
Spatial Holdout



RMSE

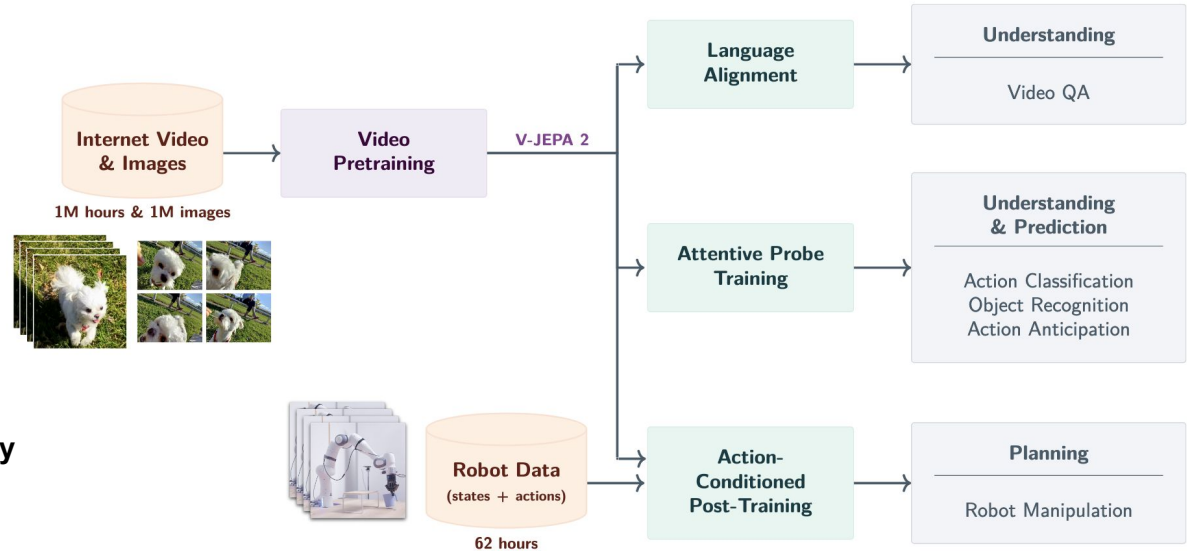


MAPE



$$\frac{\sum \frac{|A - F|}{A} \times 100}{N}$$

V-JEPA2 Model - Predicting LFMC% via NAIP Images



Using V-JEPA 2 for aerial imagery

Self-supervised learning

Large vision model → low-dim features

Assran, M., Caron, M., Misra, I., Chen, T., Li, J., Kirillov, A., Goyal, P., & Joulin, A. (2024). *V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning*. arXiv. <https://arxiv.org/abs/2406.07659>

V-JEPA2 Model - Predicting LFMC% via NAIP Images

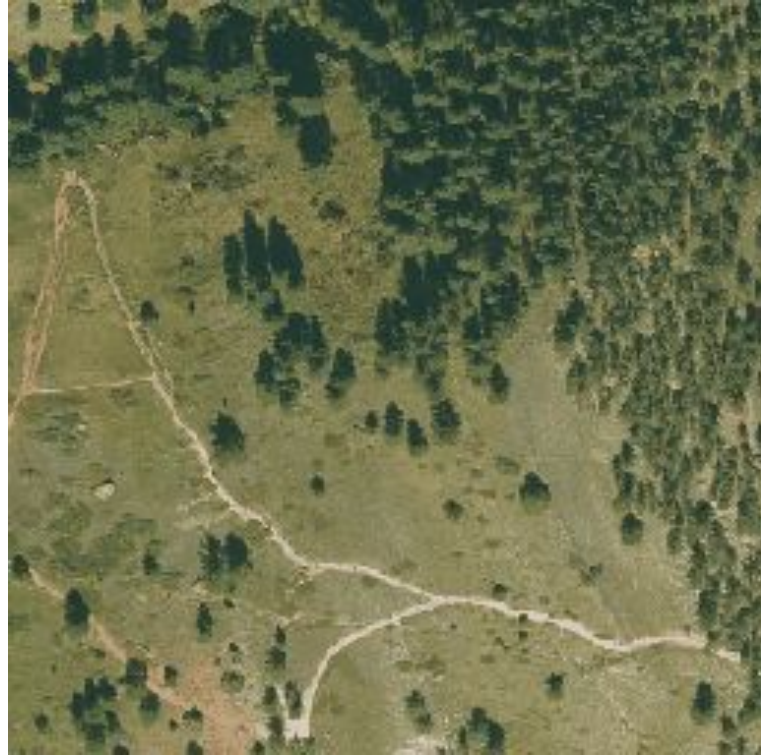
NAIP = high-res aerial image

General model applied on geospatial data

Embedding landscape structure

Simulating time via spatial windows

Frame = moving patch across space



V-JEPA2 Model - Predicting LFMC% via NAIP Images

Similar land → similar embeddings

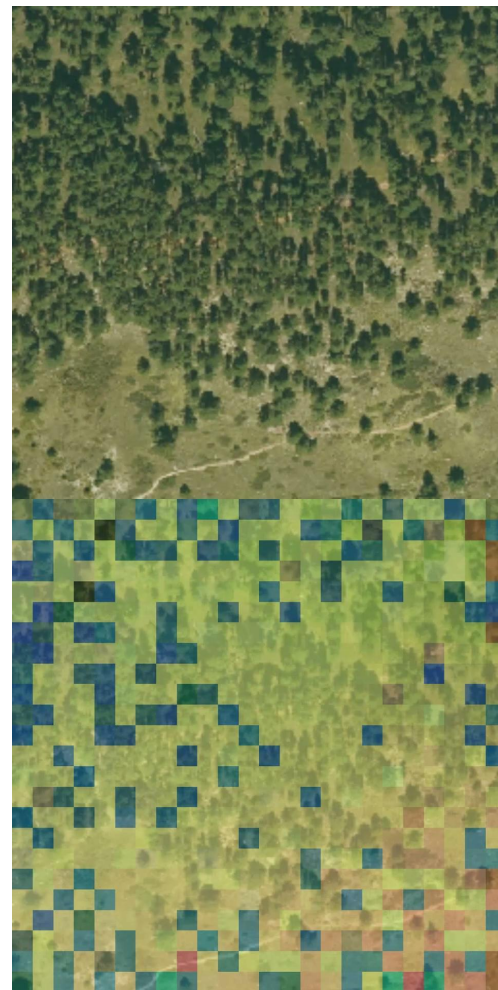
Proof of concept successful

General vision → geospatial insight

Goal: predict LFMC from imagery

Small dataset + pretrained large model = understanding

Promising path for data-scarce modeling



Shared Vision

- **Shared goal: leverag large general-purpose models to enhance geospatial understanding**
- **Models like Grid4D, AlphaEarth, V-JEPA encode foundational understanding of the world**
- **Combine global knowledge base with our domain-specific data, to create more powerful, more generalizable models, especially in data-scarce environments**
- **Shift from building alone to collaboration across disciplines and institutions, share foundational models, and then adapt and fine-tune them for specialized use**
- **Move toward a world where models can be mergeable, and synergistic—our work would be far stronger together than apart**